a How each model scores an image for finding f

Released models, frozen; shown with the registered prompt template T1

INPUT	IMAGE ENCODER	TEXT SIDE AND COMPARISON	SCORE
Radiograph 224 x 224 px	**CheXzero** CLIP ViT-B/32 image embedding	**Prompt pair** "{f}" and "no {f}", CheXzero text encoder; cosine similarity with each	**Score** Softmax over the two similarities; mean over 10 checkpoints
Radiograph 448 x 448 px	**CLEAR** DINOv2 ViT-B/14 (with registers) 768-dimensional image embedding	**Direct route** "{f}" and "no {f}", CLEAR text encoder; cosine similarity with each **Concept route** Similarity to each of 368,294 concepts weights its SFR-Embedding-Mistral vector (4,096-d); the normalised sum is compared with "{f}" and "no {f}"	**Score** Softmax over the two similarities, for each route

b Study design

1 Oct 2026 — Protocol registered on OSF, before any performance metric was computed

3 Oct 2026 — Amendments 1 and 2 filed on OSF, before their new analyses were run

MODELS, PROMPTS AND RULES

Models (frozen, zero-shot)

CheXzero, mean of 10 checkpoints ●
CLEAR direct route ▲
CLEAR concept route ■
Scoring as in panel a

Prompt templates

T1 (registered):
{f} / no {f}
Added in Amendment 1:
T2 chest x-ray showing {f} / chest x-ray showing no {f}
T3 there is {f} / there is no {f}
T4 findings consistent with {f} / no evidence of {f}

Eligibility rule (registered)

A finding enters a subgroup contrast only with at least 30 positive images from at least 10 patients per group
NIH: 14 findings for sex, 11 for age

DATASETS

NIH ChestX-ray14 — Registered

Official test split: 25,592 images, 2,797 patients, 14 text-mined labels
Sex: 1,240 female, 1,557 male patients
Age: 0-18, 18-40, 40-60, over 60 years
Held-out training patients: 2,800 (thresholds, concept selection)
Patient bootstrap, 1,000 resamples

Indiana University collection

7,466 images, 3,851 reports
Problems-field labels; report bootstrap
Run before Amendment 1, disclosed in it

VinDr-CXR — Amendment 2

15,000 training images, 13 findings
Majority of 3 radiologists
Image bootstrap; DICOM sex and age

ANALYSES

Primary analysis (NIH) — Registered

Macro-AUROC over 14 findings
Female minus male (95% CI)
Age bands minus 40-60 (98.33% CI)
Secondary: posteroanterior only; FNR

Prompts and concepts — Amendment 1

NIH: prompt robustness, T1 to T4
Concept relevance, single-concept probe, representation spread

External comparison

CLEAR concept minus CheXzero, macro-AUROC
Rule fixed before running: consistent with the CLEAR report, inconsistent, or inconclusive
VinDr-CXR: sex and age subgroups (exploratory)